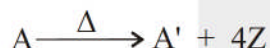
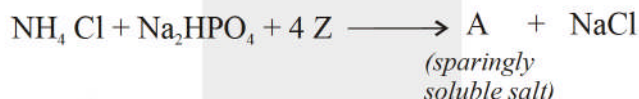


WORK SHEET - 03

SALT ANALYSIS

Comprehension

Question No. 1 to 3 (3 Question)

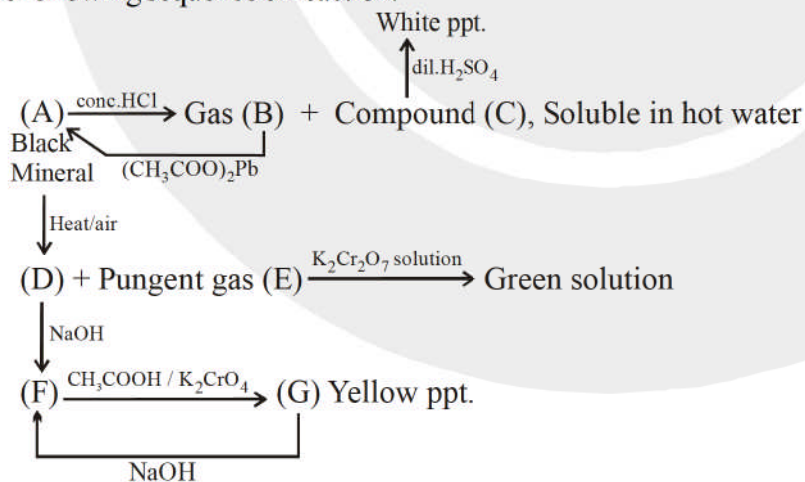


Compound (Z) is a neutral oxide and turns anhydrous CuSO_4 blue. Gas (Y) is a basic gas which can also be obtained by treating ammonium sulphate with lime water. Sequences mentioned above involve balanced reactions.

- Q.1 The compound 'X' is
(A) NaHPO_4 (B) Na_3PO_3 (C) NaPO_3 (D) Na_3PO_4
- Q.2 Which of the following reactions would produce gas 'Y'?
(A) $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \longrightarrow$ (B) $\text{Zn} + \text{NaOH} + \text{NaNO}_3 \xrightarrow{\Delta}$
(C) $(\text{NH}_4)_2\text{CO}_3 \xrightarrow{\Delta}$ (D) All of these
- Q.3 Formula of (A) is :
(A) $\text{Na}_2(\text{NH}_4)\text{PO}_4 \cdot 4\text{H}_2\text{O}$ (B) $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
(C) $\text{Na}(\text{NH}_4)_2\text{PO}_4 \cdot 4\text{H}_2\text{O}$ (D) $(\text{NH}_4)_2\text{HPO}_4 \cdot 4\text{H}_2\text{O}$

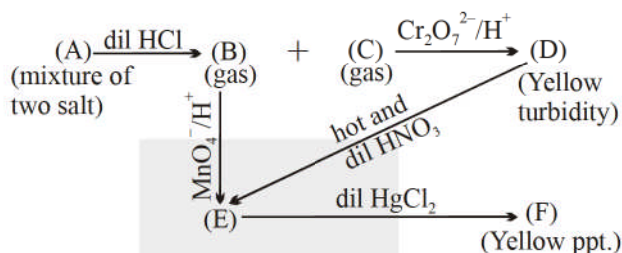
Question No. 4 to 6 (3 Question)

Read the following sequence of reaction.



- Q.4 The number of electron gained by Cr atom when gas (E) is passed through it
(A) 1 (B) 3 (C) 2 (D) 4
- Q.5 Compound (F) is
(A) PbO (B) PbS (C) PbSO_4 (D) Na_2PbO_2
- Q.6 The compound (G) is
(A) $\text{Pb}(\text{OH})_2$ (B) PbCrO_4 (C) PbCrO_2 (D) $\text{Cr}(\text{OH})_3$

Question No. 7 to 9 (3 Question)



Q.7 Find the anion(s).

- (A) SO_3^{2-} (B) $\text{SO}_3^{2-}, \text{S}^{2-}$ (C) $\text{SO}_3^{2-}, \text{CO}_3^{2-}$ (D) $\text{S}_2\text{O}_3^{2-}$

Q.8 Find out (E)

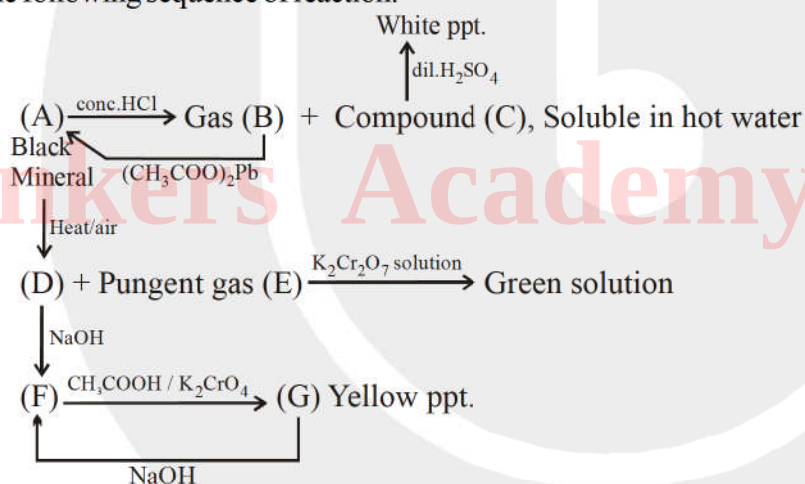
- (A) S^{2-} (B) CO_3^{2-} (C) $\text{S}_2\text{O}_3^{2-}$ (D) SO_4^{2-}

Q.9 Find out (F)

- (A) $\text{HgSO}_4 \cdot 2\text{HgO}$ (B) $\text{HgSO}_4 \cdot 3\text{HgO}$ (C) HgSO_4 (D) $\text{Hg}_2\text{SO}_4 \cdot 3\text{HgO}$

Question No. 10 to 12 (3 Question)

Read the following sequence of reaction.



Q.10 Gas (B) on passing through CdSO_4 solution will give

- (A) Black ppt. (B) Yellow ppt. (C) Orange ppt. (D) No ppt.

Q.11 Compound (A), (B) and (E) are respectively.

- (A) $\text{CuS}, \text{H}_2\text{S}, \text{SO}_2$ (B) $\text{PbS}, \text{H}_2\text{S}, \text{SO}_2$
 (C) $\text{PbS}, \text{H}_2\text{S}, \text{SO}_3$ (D) $\text{ZnS}, \text{H}_2\text{S}, \text{SO}_2$

Q.12 Compound (C) and (D) are respectively

- (A) $\text{PbO}, \text{PbCl}_2$ (B) $\text{PbCl}_2, \text{PbO}$
 (C) PbO, PbO_2 (D) PbS, PbO

Question No. 13 to 15 (3 questions)

Compound 'A' $\xrightarrow{\Delta}$ Initially swelled $\xrightarrow[\text{strong heating}]{\Delta}$ Amorphous powder

$\swarrow$
Lilac flame in the flame test.

Compound 'A' $\xrightarrow{\text{excess NaOH}}$ 'B' (No change in colour) $\xrightarrow{\text{H}_2\text{O}_2}$ 'C' (Yellow solution) (in aq. solution)

- Q.13 Compound 'A' is having water of crystallization by the number of
(A) 10 (B) 20 (C) 24 (D) 36
- Q.14 The compound 'B' is having oxidation state of
(A) zero (B) II (C) III (D) IV
- Q.15 The hybridization of compound 'C' is
(A) sp^3 (B) sp^3d (C) d^2sp^3 (D) d^3s

Match the column

Q.16 **Column I**

- (A) $\text{ZnCl}_2 + \text{H}_2\text{S}$
(B) $\text{CuSO}_4 + \text{Excess KI}$
(C) $\text{Pb}_3\text{O}_4 + \text{conc. HNO}_3$
(D) $\text{FeCl}_3 + \text{H}_2\text{S}$

Column II

- (P) Pale green colouration
(Q) Brown ppt. / appearance
(R) White turbidity
(S) No change is observed

Q.17

Column I

Column II

- (A) Cation + $(\text{NH}_4)_2\text{S} \rightarrow$ Metal sulphide $\downarrow$ $\xrightarrow{\text{Hot \& dil. HNO}_3}$ Soluble (Black) (P) Pb^{2+}
- (B) Cation + $\text{Na}_2\text{S}_2\text{O}_3 \rightarrow$ Metal thiosulphate $\downarrow$ $\xrightarrow{\text{Excess hypo soln.}}$ Soluble (white) (Q) Cu^{2+}
- (C) Cation + $\text{KCN} \rightarrow$ Metal cyanide $\downarrow$ $\xrightarrow{\text{Excess KCN}}$ Soluble (white) (R) Hg^{2+}
- (D) Cation + $\text{KI} \rightarrow$ Metal iodide $\downarrow$ $\xrightarrow{\text{Excess KI}}$ Soluble (coloured) (S) Ag^+

Q.18 Which of anions in the Column-I shows one or more observations from the column -II.

Column I

Column II

- (A) S^{2-} (P) White ppt. with AgNO_3
- (B) NO_2^- (Q) Evolution of pungent smell gas with (Al + conc. NaOH)
- (C) SO_3^{2-} (R) Brown fumes with conc. H_2SO_4 (hot)
- (D) CH_3COO^- (S) Decolourises acidified KMnO_4

Q.19	Column-I Acid redicals	Column-II Observation obtained from the acid radical
(A)	$C_2O_4^{2-}$	(P) Purple colour of acidified $KMnO_4$ is decolorised
(B)	$S_2O_3^{2-}$	(Q) Violet ppt is obtained with $[Ni(en)_3](NO_3)_2$
(C)	S^{2-}	(R) White ppt. is obtained on addition of $BaCl_2$ solution
		(S) Colourless gas is evolved on addition of dil. H_2SO_4

Q.20	Column I (Reagent)	Column II (Gives ppt / complex with)
(A)	$K_4[Fe(CN)_6]$	(P) Zn^{2+}
(B)	$K_3[Fe(CN)_6]$	(Q) Cd^{2+}
(C)	KCN	(R) Cu^{2+}
		(S) Ag^+

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ANSWER KEY

Q.1	C	Q.2	D	Q.3	B	Q.4	B	Q.5	D	Q.6	B	Q.7	B
Q.8	D	Q.9	A	Q.10	B	Q.11	B	Q.12	B	Q.13	C	Q.14	C
Q.15	D												
Q.16	(A) S; (B) Q, (C) Q; (D) P, R					Q.17	(A) P,Q,S, (B) P,Q,R,S, (C) Q,S, (D) P,R						
Q.18	(A) S; (B) P,Q,R,S; (C) P,S; (D) P					Q.19	(A) P, R ; (B) P, Q, R, S ; (C) P, S						
Q.20	(A) PQRS (B) PQRS (C) PQRS												

